Problem Set 2

Due on Canvas at noon Friday October 1st

Instructions: **No partial credit will be awarded.** All answers must be entered in the Problem Set 2 Quiz on Canvas. Note that it is not possible to edit your responses on Canvas after submitting. **Best practice is to complete the problem set on paper and then transfer solutions to Canvas.** You may collaborate on this problem set with other students in your learning team, but each student must submit his or her own solutions.

Grading: Grading is done on a ten point scale across all problems in the problem set. **Note that after submitting on canvas, you will see a score, but that does not include any points for any qualitative questions**. Grades will be finalized a few days after the due date.

Material covered: This problem set covers lectures 5-9.

Problem 1. Avocados

a. In 1980, less than 1% of avocados in the US were imported. California supplied the vast majority of avocados grown in the US. The supply of avocados (in millions of pounds) from California is given by:

$$Q_S = \frac{1}{2}(250P - 100)$$

Demand for avocados (in millions of pounds) in the US is given by:

$$Q_D = 400 - 100P$$

If the US only has access to California avocados, what would the equilibrium price (for a pound of avocados) and quantity be?

Price: ___________________

Quantity: ___________________

$$\frac{1}{2}(250P - 100) = 400 - 100P$$
$$225P = 450$$
$$P = 2$$
$$Q = 200$$

b. By 2019, avocado consumption had increased and most avocados consumed in the US were imported.[1] Mexico is by far the largest producer of avocados in the world. Given the favorable climate, Mexico is able to supply avocados at a price of $1 per pound (and there is effectively no limit on how many can be grown).

If there is free trade with Mexico, how many millions of pounds of avocados will be consumed in the US, and what share of them will be produced in the US? Are US producers and consumers better or worse off from trade?

Quantity consumed: ___________________

US-produced Share: ___________________

☐ US consumers and US producers are better off

☐ US consumers better, US producers worse off

☐ US consumers worse, US producers better off

[1] The true figure is approximately 90%.

☐ US consumers and US producers worse off

$$Q_D(P=1)=400-100=300$$
$$Q_S(P=1)=75$$
$$Share=25\%$$
$$Imports=225$$
$$US\ producers\ worse, consumers\ better$$

c. The government is considering imposing a tariff on imported avocados to help US farmers. Suppose they were to levy a tariff of $0.20 per pound on imports. By how much would US consumer surplus increase or decrease from this tariff? By how much would US producer surplus increase or decrease from this tariff? How much government revenue would the tariff generate? (Hint: draw a diagram of the situation and compute areas corresponding to the different objects at the free-trade equilibrium and tariff equilibrium).

Change in US consumer surplus from tariff: ___________________

Change in US producer surplus from tariff: ___________________

Government revenue from tariff: ___________________

Answer. Price will increase to $1.20, as this is still less than the no-trade price. The new equilibrium has $Q_D=280$, $Q_S=100$. Consumer surplus goes from $(0.5)(3)(300)=450\ to\ (0.5)(2.8)(280)=392$, for a change of -58. Producer surplus goes from $(0.5)(0.6)(75)=22.5\ to\ (0.5)(0.8)(100)=40$, for a change of $+17.5$. Tariff revenue would be $0.2 times the quantity imported (which is quantity demanded less domestic supply), or $(180)(0.2)=36$.

d. Avocado lovers are not happy about the plan and decide to lobby against it. The government proposes to hand back 100% of the tariff revenue it collects to avocado consumers at the end of the year.[2] Would this make avocado lovers just as well-off as under free trade? Why?

☐ Yes

☐ No

☐ Need more information

Why? ____

____

[2] Ignore the practical challenges of doing this for now; assume they have a way to do this.

____

____

Answer. No, it would not, as the tariff revenue (36) is less than the change in consumer surplus (-58). This is the key insight: even total surplus falls from the tariff.

Problem 2. *E-Scooters.* The e-scooter craze is sweeping the nation, with cities like Los Angeles, Chicago, and Atlanta overrun with dockless electric scooters that can be rented via an app. A New Haven startup, Scüt, is hoping to bring the craze to the Elm City. They have determined that their daily cost function (where quantity is in scooters placed around the city in a given hour of the day) is

$$TC(q) = 40 + q + 0.1q^2$$

Unfortunately for Scüt, the e-scooter business is perfectly competitive, and they may face rivals such as Lime, Bird, and even Uber and Lyft in the future.

(a) What is Scüt's supply function for scooters (for the range of prices where they are willing to supply any), **as a function of the market price p (which is daily revenue per scooter) for an e-scooter**?

Supply function: $Q_S(P) =$____

$\boldsymbol{MC = 1 + 0.2q}$

$$\boldsymbol{q = 5P - 5}$$

(b) If the current prevailing market price (hourly revenue) in the short run is \$7 per hour per scooter, how many scooters will Scüt supply, and what will their hourly profit be?

Supplied Quantity: $Q_S(P) =$___________ **30** ______________________

Hourly Profit (to 2 decimal points): __________ 30*7-40-30-0.1*(30*30)=50 __________

____

(c) Scüt's success has attracted some attention, leading to a price war in New Haven. In the short-run, the fixed cost component of total costs is unavoidable. What is the price below which Scüt would stop placing scooters around the city in the short term?

Price $1 (would accept $1.1)

$$\boldsymbol{AVC = 1 + 0.1q}$$

(d) In the long run, there is free entry, and indeed several competitors join the market with the same cost function as Scüt. What will be the long-run equilibrium price (hourly revenue) per e-scooter?

Price **5**

$$ATC = \frac{40}{q} + 1 + 0.1q$$
$$MC = 1 + 0.2q$$
$$\frac{40}{q} + 1 + 0.1q = 1 + 0.2q$$
$$40 = 0.1q^2$$
$$400 = q^2$$
$$q = 20$$
$$MC = 5$$

Problem 3. Car Pricing

Hyundai is developing an all-new, premium, low-cost electric car to compete with Tesla's Model 3. They plan to market the vehicle in both the US ("U") and Asia ("A"). Demand is estimated to be

$$Q_U = 4{,}000{,}000 - 100P_U$$
$$Q_A = 6{,}000{,}000 - 200P_A$$

They estimate that they will be able to produce the car at a constant marginal cost of $25,000 in either location.

a) First, assume it is very costly to ship cars across the Pacific, and so the company is able to charge different prices in different parts of the world. What is the optimal price, quantity, and profit level in each market?

US Price: ________________

US Quantity: ________________

US Profit: ________________

Asia Price: ________________

Asia Quantity: ________________

Asia Profit: ________________

$$US: P_U = 40{,}000 - 0.01Q_U$$
$$MR = 40{,}000 - 0.02Q$$
$$0.02Q = 15000$$
$$Q = 750{,}000$$
$$P = 32{,}500$$
$$\pi = \big((32500 - 25000) * 750000\big)$$
$$= 5{,}625M$$

ASIA

$$P_A = 30{,}000 - 0.005Q_A$$
$$MR = 30{,}000 - 0.01Q$$
$$25000 = 30000 - 0.01Q$$
$$0.01Q = 5000$$
$$Q = 500{,}000$$
$$P = \$27{,}500$$
$$\pi = (2500 * 500000) = 1{,}250M$$

b) The company is concerned is very concerned about its image in the US and is worried about a backlash from consumers if they charge more in one market than another. What is the optimal price, quantity and profit level if they are constrained to charge the same price in both markets? (Note, answers may not be round numbers – you may round price to the nearest \$100 and profit to the nearest \$1M).

$$Q = 10{,}000{,}000 - 300P$$
$$P = 33{,}333.33 - 0.00333Q$$
$$MR = 33{,}333.33 - 0.006666Q$$
$$25000 = 33{,}333.33 - 0.0066666Q$$
$$8333.33 = 0.006666q$$
$$q = 1.25M$$
$$p = \$29{,}167$$
$$\pi = (4167 * 1250000) = 5{,}209M$$

BUT – this is worse than just serving the US market! (e.g. the optimal price is above the kink in the demand curve)

SO – charge $32,500 everywhere, have no sales in Asia, make 5.625B, sell 750,000

c) The company has determined that they could pay $100M to make the US version sufficiently different, with an entirely different marketing campaign, such that they could charge different prices in the two different markets. Is this a good idea for the firm to do?

- ☐ Just charge the same price everywhere
- ☐ Charge different prices in Europe and the US, incur $100M cost
- ☐ Not sure / Not enough information.

Answer: Yes, being able to earn $1.250B in the Asian market is worth $100M